This article is now in press at *Aging, Cognition, and Neuropsychology*.

Loneliness and social isolation are not associated with executive functioning in a cross-sectional study of cognitively healthy older adults

Katelyn S. McVeigh[a*], Matthias R. Mehl[b], Angelina J. Polsinelli[c,d], Suzanne A. Moseley[e], David A. Sbarra[f], Elizabeth L. Glisky[g,h], and Matthew D. Grilli[a,h,i]

[a] Human Memory Lab, Department of Psychology, University of Arizona, Tucson, Arizona, USA

[b] Naturalistic Observation of Social Interaction Laboratory, Department of Psychology, Tucson, Arizona, USA

[c] Department of Neurology, Indiana University School of Medicine, Indianapolis, Indiana, USA

[d] Indiana Alzheimer's Disease Research Center, Indianapolis, Indiana, USA

[e] Minnesota Epilepsy Group, St Paul, Minnesota, USA

[f] Laboratory for Social Connectedness and Health, Department of Psychology, University of Arizona, Tucson, Arizona, USA

[g] Aging and Cognition Laboratory, Department of Psychology, University of Arizona, Tucson, Arizona, USA

[h] Evelyn F. McKnight Brain Institute, University of Arizona, Tucson, Arizona, USA

[i] Department of Neurology, University of Arizona, Tucson, Arizona, USA

*Corresponding author:
Katelyn S. McVeigh, MA
1503 E. University Blvd
Tucson, AZ, 85721
kkmcveigh@arizona.edu

FUNDING: This work was supported by Mind and Life Institute (2015-1440-Polsinelli)

Abstract

The empirical literature on the relationship between social interaction and executive functions in older age is mixed, perhaps stemming in part from differences in executive functioning measures (e.g., a single test or a test battery) and differences in conceptualization and measurement of social interaction (e.g., loneliness and/or social isolation). Our study investigated the relationship between social interaction and executive functions in a sample (N = 102) of cognitively unimpaired, functionally independent older adults (ages 65 - 90). Participants received a comprehensive executive functioning battery with measures of working memory, inhibition, shifting, and global executive functioning. We measured loneliness subjectively through survey (NIH Toolbox Loneliness Scale) and social isolation objectively through naturalistic observation (percentage of time alone captured from ambient sound sampling using the Electronically Activated Recorder). Loneliness was not significantly related to working memory, inhibition, shifting, or global executive functioning (p-values = .13 - .65). Time spent alone was also not significantly related to working memory, inhibition, shifting, or global executive functioning (p-values = .11 - .69). Corresponding Bayes factors indicated anywhere from moderate to extremely strong evidence (BF_{01} = 8.70 to BF_{01} = 119.49) in support of no relationship and the effect sizes were at best modest (β values = -.17 - .10). These findings suggest that, among cognitively healthy older adults, there may not be a robust cross-sectional relationship between multiple aspects of executive functioning and either subjectively experienced loneliness or objectively observed social isolation.

Keywords: loneliness, social isolation, social interaction, executive functioning, executive functions, aging, older adults, cross-sectional, cognitively healthy

Introduction

Social interaction is theorized to have a close connection to cognition, particularly among older adults (Fratiglioni et al., 2004). Although socializing is thought to be related to multiple cognitive domains (Wascher et al., 2018), executive functions, which include inhibition, working memory, set-shifting, and cognitive flexibility (Diamond, 2013), are considered particularly important (Madjar et al., 2019; von Hippel, 2007). One perspective is that executive functions may facilitate meaningful and complex social interactions. For example, theory of mind, meaning the ability to understand another person's mental state and predict their behavior (Premack & Woodruff, 1978), is believed to be important for social interactions (Krendl et al., 2022) and possibly supported by executive functions (Aboulafia-Brakha et al., 2011; Otsuka et al., 2021). This could reflect that theory of mind relies on shifting perspectives, maintaining focus on the other individual, and considering multiple pieces of information (Roca, 2016). Executive functions also could help form and maintain social connections by enabling an individual to make social plans and remember important details that are relevant to future social interactions (e.g., details from a prior conversation) (Anderson et al., 2001; Delis et al., 2001; Mejia et al., 1998). Another, and perhaps non-mutually exclusive, perspective is that social interaction may promote the use and preservation of executive functions (Perry et al., 2019). In other words, maintaining a socially engaged lifestyle may inherently promote the complex use of various aspects of executive functioning, which in turn may strengthen, or at least maintain, their effectiveness more generally. Given the potentially bidirectional relationship between social interaction and executive functions, it is not surprising that this topic has received attention in recent years, particularly in an aging context, as the field aims to better understand what factors might place cognitively healthy older adults at increased risk for cognitive decline.

Despite the theorized connection between social interaction and executive functions among older adults, evidence supporting the association is mixed (DiNapoli et al., 2014; Gilmour, 2011; Griffin et al., 2020; Krueger et al., 2009; Montoliu et al., 2019; O'Luanaigh et al., 2012; Schnittger et al., 2012; Shankar et al., 2013, Wilson et al., 2007). Although some studies find significant relationships, others do not, especially after controlling for factors such as age, education, and depression. Relatedly, a recent meta-analysis of 51 longitudinal studies including middle-aged and older adults suggests that social interaction, broadly measured, may be weakly (albeit significantly) related to future executive functioning, broadly defined (e.g., r = .03 in Evans et al., 2019). The equivocal findings could partly stem from variability in methodology. Many basic study parameters have differed in significant ways, including sample size (DiNapoli et al., 2014; Gilmour, 2011), and whether individuals with cognitive impairment or medical conditions were included (DiNapoli et al., 2014) or not (Krueger et al., 2009; O'Luanaigh et al., 2012). The research designs are mixed as well, with some longitudinal studies (Zhong et al., 2017; Shankar et al., 2013; Wilson et al., 2007) and others being single timepoint or cross-sectional investigations (DiNapoli et al., 2014; O'Luanaigh et al., 2012; Krueger et al., 2009). Additionally, the cognitive tests used to measure executive functions and its corresponding subdomains vary in the number and type of tests used. Of note, the type of executive functioning tasks used and broader study parameters, such as sample size and design, are not clearly linked to study outcomes. For instance, studies that found evidence of a relationship between social interaction and executive functioning ranged widely in sample size (i.e., 267 to 14,199), as did studies that found no relationship (i.e., 86 to 13,176).

Whether part of the ambiguity in outcomes reflects how social interaction is measured is unclear. Both loneliness and isolation are implicated in adverse health outcomes, including

cognitive decline (Evans et al., 2019), as well as functional decline, depression, and anxiety (Perissinotto et al., 2012; Shankar et al., 2013). In fact, loneliness and social isolation are believed to be two features of social interaction that capture unique, but often intertwined, experiences (Newall and Menec, 2019; Perlman and Peplau, 1998). One model by Cacioppo and Cacioppo (2018) proposes that older adults who feel lonely begin to perceive their social environments as dangerous or threatening, which leads them to have and recall more negative experiences (Cacioppo & Cacioppo, 2018). This results in older adults either further disengaging from their social networks, or their social networks begin to disengage from them. On the other hand, Barnes and colleagues' theoretical framework proposes a bidirectional relationship between social isolation and loneliness, taking into account pre-existing risk factors (e.g., age, living alone, depression, anxiety, chronic disease, etc.) and relationships to other health outcomes (Barnes et al., 2021; NASEM, 2020). According to this model, for older adults, losing loved ones, deteriorating health, and decreasing network size could increase their risk of experiencing loneliness and/or isolation (Dykstra et al., 2005), which may contribute to adverse health outcomes, including cognitive decline.

Despite their interconnectedness and common hypothesized relevance to understanding cognition, there are conceptual and methodological differences between loneliness and social isolation. Loneliness is defined as a subjective, distressing feeling accompanying a perceived lack of social and emotional support, whereas social isolation is an objective index of social interaction defined by parameters such as the time spent around others, network size, and frequency of social contact (de Jong Gierveld & Havens, 2004). Reasonably, loneliness is often assessed via participants' self-reports (often using the UCLA Loneliness Index, De Jong Gierveld Loneliness Scale, or even single-item measures). However, self-report questionnaires

are also commonly used to measure social isolation, often asking participants to report on their social network size and the frequency of social interaction using measures similar to the Social Network Index (Cohen et al., 1997). Although self-report measures are valuable for gathering insight into people's perceived experiences, they are subject to error or biases that can affect accuracy of identifying observable behaviors (Althubaiti, 2016; Raina et al., 2002). Social desirability bias, for instance, occurs when participants are motivated to overreport socially desirable behaviors/attitudes and underreport undesirable behaviors/attitudes (Latkin et al., 2016), perhaps to avoid negative outcomes from disclosing sensitive information (Tourangeau & Yan, 2007). If, for example, participants perceive being lonely/isolated as undesirable and anticipate embarrassment or feeling judged as a result, they may reluctantly or not accurately report on their social networks, time spent around others, or feelings of loneliness.

In sum, mixed results in the literature to date have left no clear expectation of if, when, or how social interaction relates to executive functions among older adults. The present study aimed to provide clarity at a cross-sectional level, and among cognitively normal older adults, by addressing two unresolved issues. First, although executive functioning is not consistently connected to social interaction, prior studies generally have not comprehensively evaluated the domains of executive functions, instead opting for single tasks or a few measures that may be incomplete or inconsistent in the theorized domains of executive functioning (i.e., updating, switching, and inhibition, Miyake et al., 2000; Miyake and Friedman, 2012). Therefore, it is unclear whether there is a nuanced relationship between social interaction and a particular subtype of executive functions, or a more global measure accounting for multiple domains of executive functioning. We attempted to close this gap by examining a comprehensive executive function battery, comprised of the types of measures of working memory, shifting, inhibition,

and overall executive functions proposed by Miyake and colleagues (Miyake at al., 2000). Second, although prior work finds mixed results in regard to isolation, the reliance on subjective report is potentially problematic, especially given that subtle memory difficulties are not uncommon even among healthy older adults, and potentially undermines the validity of reports of how much time they spend alone. Whether an observational measure of isolation would reveal evidence for, or against, a relationship to executive functioning is not yet known. To address this point, we used the Electronically Activated Recorder, or EAR (Mehl et al., 2001; Mehl, 2017), a mobile-app based unobtrusive, naturalistic observation method that records ambient audio data from participant's daily lives. Using the EAR, we were able to quantify, based on the sampled ambient sounds, how often individuals spent time alone, which constitutes an objective measure of social isolation (Milek et al., 2018). As a self-report comparison, and to measure loneliness, we used the loneliness scale from the National Institutes of Health (NIH; Cyranowski et al., 2013).

In light of the methodological limitations discussed, we considered it an empirical question whether certain aspects of executive functioning (shifting, inhibition, or working memory) would be more strongly related to objectively measured social isolation or subjectively measure loneliness among cognitively normal older adults. As noted, social isolation might relate to executive functioning due to the potential importance of social interaction for fostering higher-order cognitive functions, or through the role of executive functions in engaging social environments. Likewise, loneliness might relate to executive functioning because of the impact of negative psychosocial connectedness or interaction on cognitive functioning, or because of the importance of executive functioning for making effective social connections, and mitigating loneliness.

Methods and Materials

Methods

We ran secondary analyses on a large dataset combined from two previous studies that examined cognition among healthy older adults (Moseley, 2018; Polsinelli, 2017; Polsinelli, et al., 2020). This study was not pre-registered, and we did not conduct an a priori power analysis. Sensitivity analyses indicate that the current sample size would be powered at 80% to detected an f^2 value of .13, which is small to medium (see Lakens, 2022 for a detailed discussion of sample size justifications when using secondary data). Participants provided written informed consent for the original studies in accordance with the Institutional Review Board of the University of Arizona. They also provided consent to have their data (including audio files) used for future studies.

Participants

Participants included 102 healthy, cognitively unimpaired, functionally independent, community dwelling older adults (53 females, 49 males, 99% White, non-Latino/a, 63% married) between the ages of 65 and 90 (M = 75.89, SD = 5.84) recruited through the greater Tucson community via community events, flyers, email listservs, and research databases at the University of Arizona. Exclusion criteria for both original studies were a) a diagnosed neurological disorder, b) the experience of a head injury with loss of consciousness for longer than 5 minutes, c) a reported history of alcohol or drug abuse, d) a reported history of significant psychiatric illness, e) current use of psychotropic medication, and f) scores of 21 or higher on the Center for Epidemiological Studies-Depression Scale (CES-D). Cognitive status was examined with the Mini Mental Status Examination (MMSE), where all participants scored 25 or better, out of a total possible score of 30 (M = 28.80, SD = 1.16). Roughly half (n = 45) of the sample

was originally recruited to examine correlations between cognition and hearing loss in older adults (Moseley, 2018) and therefore has a wider range of hearing abilities. Existing literature suggests an association between hearing loss and higher levels of loneliness (Chen, 1994; Sung et al., 2016). However, our participants did not report very high levels of loneliness (hearing-loss sample: $M = 8.30$, $SD = 3.56$) and there was not a significant difference in loneliness between the two samples, $t(97) = 0.124$, $p = 0.902$. There was also not a significant difference in social isolation between the samples, $t(97) = 0.927$, $p = 0.356$.

Materials and Procedures

Participants for both original studies came into the laboratory to complete three hours of cognitive testing (two sessions separated by a week), including the executive functioning measures and the NIH Loneliness questionnaire used in the present study. In the week between testing sessions, the participants wore the Electronically Activated Recorder (EAR).

Executive Functions

Executive functions were measured with a test battery that allowed us to assess working memory, inhibition, shifting, and global executive functioning (see Table 1). Miyake and colleagues (2000) had previously identified these tests as representing three independent, but related aspects of executive functioning. Working memory, the ability to temporarily store and manipulate information, was measured with Keep Track and Consonant Updating. Inhibition, the ability to suppress a pre-potent behavioral response in favor of another response, was measured with Stroop. Shifting, the ability to switch back and forth between tasks, was measured with Global-Local and Letter-Number. Each task, except the Stroop task, was completed on a computer in the laboratory using E-prime software (Schneider, Eschman, & Zuccolotto, 2002). Raw scores on all tests were converted to z-scores. To maintain consistency with prior work

(Polsinelli et al., 2020), and to minimize the number of statistical comparisons, we used composite z-scores for overall executive functioning and the domains of working memory and shifting by averaging individual z-scores within each domain (or all tests for overall executive functioning). Correlations among executive functioning measures are in Table 2, which shows that within domain tasks tended to be most strongly correlated, relative to tasks across domains, and all tasks were significantly correlated with the overall EF composite score.

Table 1. Description of cognitive tests used to measure executive functions.

Keep Track (Miyake et al., 2000; Yntema, 1963)	Participants were shown a list of 15 words from predetermined categories (furniture, fruits, metals, sports, relatives, and distances) and instructed to hold in mind the last item shown in each category.
Consonant Updating (Miyake et al., 2000; Morris & Jones, 1990)	Participants were shown a string of consonants and instructed to say aloud the last four letters as each appeared on the computer screen. After the string of letters ended, participants were instructed to repeat the final four letters again.
Stroop (Golden, 1978)	Participants were given a page of "XXXX" printed in green, red, or blue and asked to name as many colored "XXXX" as they could. Participants were then given a page with the words "GREEN", "RED", or "BLUE" printed in an incongruent color and asked to name the color of the ink that the words were printed in.
Global-Local (Miyake et al., 2000)	Participants were shown Navon figures (Navon, 1977), which are large shapes composed of an outline of closely-spaced smaller shaped (e.g., a large square composed of smaller triangles). If the figure was black, participants were instructed to identify the larger shape. If the figure was blue, participants were instructed to identify the smaller shapes that compose the larger shape.
Letter-Number (Miyake et al., 2000; Rogers & Monsell, 1995)	Participants were shown number-letter pairs (e.g., 4B) in one of four quadrants on the laboratory computer screen. Participants were instructed to decide if the number was odd or even when the letter-number pair was shown in top half of the screen. Participants were instructed to decide if the letter was a consonant or vowel when the letter-number pair was shown in the bottom half of the screen.

Table 2. Correlations among executive functioning measures

Intercorrelations					EF measure	Composite type	Correlation with EF composite measure
KT	CU	GL	LN	Stroop			
	.29**	.21*	.12	.22*	Keep Track (KT)	WM	.61**
		.17	.19	.26*	Consonant Updating (CU)	WM	.64**
			.31**	.01	Global-Local (GL)	Shifting	.57**
				.21*	Letter-Number (LN)	Shifting	.61**
					Stroop	Inhibition	.58**

*p <.05, **p <.01
Note: EF = executive functioning; WM = working memory

The Electronically Activated Recorder (EAR) and Social Isolation

The EAR (Mehl et al., 2001; Mehl, 2017) application was downloaded onto an iPod Touch and programmed to randomly sample 30-second sound files from the participant's surrounding environment, every 6 to 18 minutes. Participants wore the EAR for four continuous days, two weekend days and two weekdays, resulting in an average of 361.23 sound files per participant (*SD* = 68.46, range = 93 – 447). Each sound file was later coded by a trained rater using a standardized behavioral coding system consisting of both general and study-specific EAR coding categories (https://osf.io/bva2c; for further general information on the EAR coding process see Kaplan et al., 2020). Sound files were excluded if they were determined to be invalid (i.e., participants were non-compliant in wearing their EAR device or if they were sleeping) (n = 5,038). The final data used for analyses included 31,808 valid (i.e., compliant), waking (i.e., not sleeping) sound files.

Social isolation was quantified as time spent alone, coded from the EAR-sampled ambient sounds (Mehl & Pennebaker, 2003). "Alone" was defined as "Participant is by themselves (alone); also applies if participant is surrounded by people, but not engaged in any sort of social interaction (e.g., the participant is by themselves, despite being potentially physically among/surrounded by other people, such as in a coffee shop)." Importantly, if the participant is apparently engaged in social interaction absent any talking (e.g., silently walking with another on the beach), the sound file would not be coded as alone, regardless of whether the participant engages in conversation (to the extent that there are other audible cues of another person being socially present/interacting with the participant). Dividing the number of sounds files coded as "alone" by total number of valid (i.e., compliant), waking (i.e., not-sleeping) sound files produced a percentage of time spent alone.

The EAR has been extensively validated as an ecological assessment tool used in a wide range of studies with different cohorts and has shown to be highly reliable across its objective measures, including time spent alone (Mehl, 2017). All of the raters for the current database completed training with the EAR protocol in which they demonstrated good to excellent reliability across the objective measures of the EAR prior to coding sound files for the present study (intraclass correlations ranging from .88 to .98). Further, in prior EAR research, the classification "alone" used here showed good construct validity through (cross-sectional) correlations with extraversion ($r = -.12$, $p < .01$; Tackman et al., 2020) and wellbeing ($r = -.19$, $p < .01$; Milek et al., 2018).

Loneliness

Loneliness was measured using the Loneliness Scale (Cryanowski et al., 2013) from the NIH Toolbox. It consists of 5 self-report items, with response options rated on a 5-point scale: never (1), rarely (2), sometimes (3), usually (4), and always (5). The 5 items consisted of: "I feel alone and apart from others," "I feel left out," "I feel that I am no longer close to anyone," "I feel alone," and "I feel lonely." Scores range from 5 to 25, with higher scores indicating greater feelings of loneliness.

Statistical Analyses

Eight separate multiple regression models examined the association between social interaction (loneliness or isolation) and executive functioning (working memory, inhibition, shifting, overall executive functioning). These analyses were run in a non-parametric fashion with rank-ordered outcome and predictor variables due to non-normal distributions of loneliness and shifting (De Winter, Gosling, & Potter, 2016) and to maintain analytic consistency with prior work (Polsinelli et al., 2020). All models included age, sex, hearing status (hearing loss or no

hearing loss), and years of education as covariates, with age and education rank-ordered. We also calculated Bayes factors for each model. In the standard scenario, Bayes factors measure evidence for an alternative hypothesis relative to the null hypothesis, and they reflect the likelihood that the data are to occur under the former relative to the latter. Two advantages of Bayes factors are they distinguish between an absence of evidence and evidence of absence, and their interpretation is not all-or-none (Keysers et al., 2021; Schönbrodt & Wagenmakers, 2018; Wagenmakers et al., 2018). Applied to the present study, Bayes factors can provide complementary information to null hypothesis significance testing *p*-values and quantify the degree to which the evidence favors relationships between executive functioning and social interaction among older adults against the null hypothesis, namely the absence of relationships between these constructs at a cross-sectional level. All analyses and graphs were run and created using RStudio (RStudio Team, 2021). Correlations between all variables are in Table 3.

Table 3. Correlations between all variables

Variable	1.	2.	3.	4.	5.	6.	7.	8.	9.
1. Loneliness	-								
2. Time spent alone	.36***	-							
3. Working Memory	-.09	-.02	-						
4. Inhibition	.04	-.06	.31**	-					
5. Shifting	-.10	-.07	.20*	.08	-				
6. Global EF	-.10	-.09	.77***	.55***	.65***	-			
7. Age	.03	-.03	-.16	-.05	.07	-.08	-		
8. Education	.02	.05	.14	.25*	.001	.20	.05	-	
9. Sex	-.20	-.35**	.15	.13	-.08	.14	.04	.16	-
10. Hearing Status	.06	-.14	-.04	-.09	-.05	-.06	.01	-.15	.29**

*$p < .05$, **$p < .01$, ***$p < .001$

Results

The mean loneliness score on the NIH Toolbox measure was 8.35 (*SD* = 3.62; range = 5 to 19), with a median score of 7 (IQR = 5-11) (Figure 1A). The average proportion of time spent alone was 0.67 (*SD* = 0.18; range = 0.24 to 1.00), and a median score of 0.68 (IQR = 0.54-0.80) (Figure 1B). Global executive functioning composite z-scores ranged from -1.90 to 1.04 (*M* = 0.03, *SD* = 0.59), shifting composite z-scores ranged from -3.20 to 1.14 (*M* = 0.01, *SD* = 0.82), working memory composite z-scores ranged from -1.56 to 1.76 (*M* = 0.05, *SD* = 0.80), and inhibition z-scores ranged from -2.76 to 2.51 (Figure 1C).

Multiple regression models showed that loneliness was not significantly related to working memory (*β* = -0.17 *p* = 0.13), inhibition (*β* = 0.04, *p* = 0.75), shifting (*β* = -0.05, *p* = 0.65), or global executive functioning (*β* = -0.08, *p* = 0.45) (Figure 2, and see Table 4 for full models). Corresponding Bayes factors indicated anywhere from strong to extremely strong (BF_{01} = 10.15 to BF_{01} = 119.49) evidence in favor of no relationship for the overall models including loneliness and these measures of executive functions. The R^2 values further indicated that the predictive power of the models were quite small.

Similarly, multiple regression models showed that time spent alone also was not significantly related to working memory (*β* = 0.02 , *p* = 0.88), inhibition (*β* = 0.10, *p* = 0.26), shifting (*β* = -0.17 , *p* = 0.11), or global executive functioning (*β* = -0.04 , *p* = 0.69) (Figure 2; Table 5 for full models). Corresponding Bayes factors indicated anywhere from moderate to very strong evidence (BF_{01} = 8.70 to BF_{01} = 58.33) in support of no relationship for the overall models including time spent alone and these measures of executive functions. As was the case for loneliness, the R^2 values suggested low predictive power for all models.

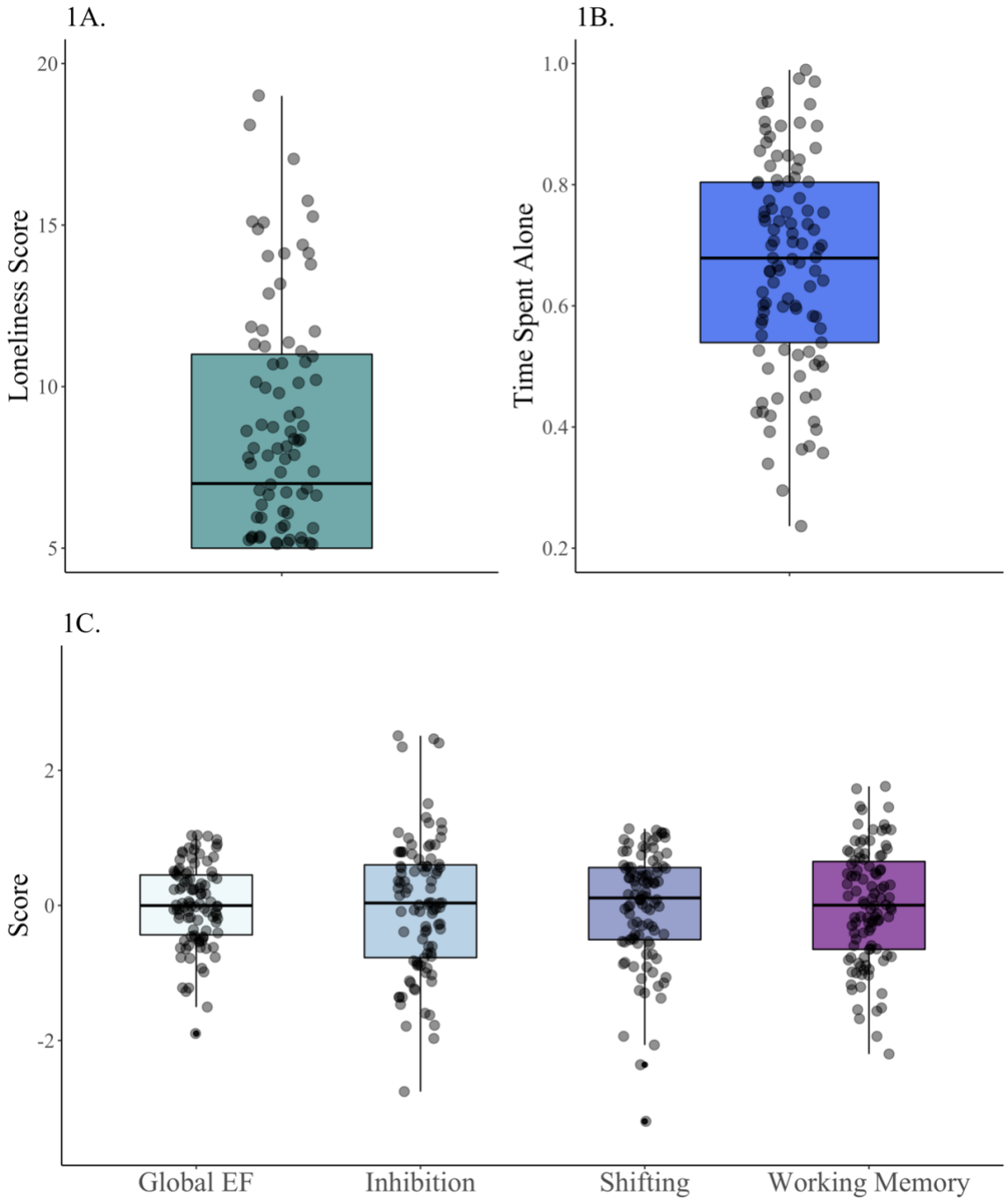


Figure 1. Distributions of loneliness scores (1A), proportion of time spent alone (1B), and executive functioning scores (1C). Z-scores for inhibition and composite z-scores for global executive functioning, shifting, and working memory.

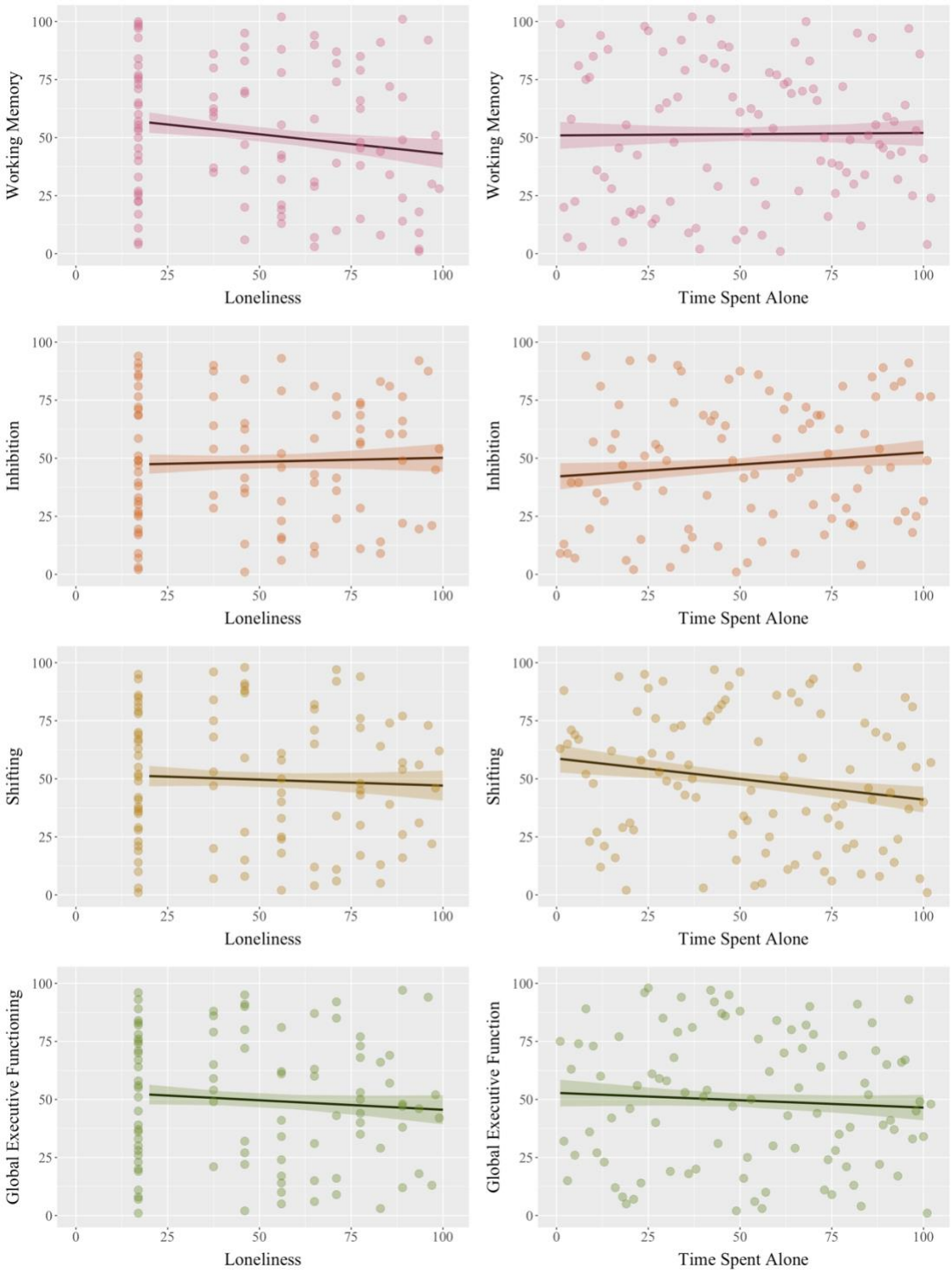


Figure 2. Scatter plots for loneliness and time spent alone (social isolation) in relation to working memory, inhibition, shifting, and global executive functioning. X and Y axis show transformed (rank ordered) data. Regression lines for each model overlaid, representing the relationship between each predictor and outcome variable, including covariates of age, education, and sex.

Table 4. Associations Between Loneliness and Executive Functions

	Working Memory			Inhibition			Shifting			Global EF		
Variable	*β*	*SE B*	*p*	*β*	*SE B*	*p*	*β*	*SE B*	*p*	*β*	*SE B*	*p*
Loneliness	-.17	.11	.13	.04	.11	.75	-.05	.11	.65	-.08	.10	.45
Age	-.21*	.10	.04	-.10	.10	.33	.01	.10	.94	-.15	.10	.14
Education	.09	.10	.39	.19	.10	.06	.02	.10	.82	.16	.10	.11
Hearing Status	-.83	6.35	.90	1.16	6.08	.85	.90	6.41	.89	.84	6.20	.90
Sex	-.27	6.00	.96	-7.08	5.73	.22	4.95	6.10	.42	1.70	5.90	.77
R^2		.03			.02			-.04			.01	
F		1.55	.18		1.34	.02		.22	.95		1.12	.40
BF_{01}		10.15			13.64			119.49			22.10	

*$p < .05$, **$p < .01$

Table 5. Associations Between Time Spent Alone and Executive Functions

	Working Memory			Inhibition			Shifting			Global EF		
Variable	*β*	*SE B*	*p*	*β*	*SE B*	*p*	*β*	*SE B*	*p*	*β*	*SE B*	*p*
Alone	.02	.10	.88	.10	.10	.26	-.17	.10	.11	-.04	.10	.69
Age	-.25*	.10	.01	-.13	.10	.20	-.01	.10	.95	-.18	.10	.07
Education	.08	.10	.43	.18	.10	.06	.04	.10	.72	.16	.10	.11
Hearing Status	2.52	6.02	.68	2.24	5.75	.70	1.84	6.01	.76	2.67	5.92	.65
Sex	-.04	6.20	.93	-4.22	5.95	.48	2.51	6.20	.69	1.24	6.10	.84
R^2		.02			.03			-.01			.02	
F		1.45	.21		1.55	.18		.75	.59		1.35	.25
BF_{01}		24.58			8.70			58.33			27.23	

*$p < .05$, **$p < .01$

Discussion

Social interaction is theorized to have important connections to executive functions. Despite considerable literature on this topic (DiNapoli et al., 2014; Gilmour, 2011; Griffin et al., 2020; Krueger et al., 2009; Montoliu et al., 2019; O'Luanaigh et al., 2012; Schnittger et al., 2012; Shankar et al., 2013, Wilson et al., 2007), evidence supporting the association is mixed. Differences in findings could stem in part from variable and limited measures of executive function. Although questionnaires are commonly used to measure loneliness, a subjective measure of perceived social and emotional support, questionnaires are also commonly used to measure social isolation, which presents an issue as isolation is an inherently objective concept. We were able to address these limitations in the present study and therefore add to the current understanding of the association between executive functions and social interaction. Participants received a comprehensive, theory-driven executive functioning battery, and we measured social isolation objectively with the EAR, a smartphone-based application used to unobtrusively and naturalistically capture isolation. We hypothesized that if executive functions and social interaction are related among cognitively healthy older adults, older adults who were lonelier or more isolated would have lower executive functions in one or more domain.

We found little support for our hypothesis. Neither loneliness or social isolation were significantly related to overall executive functions, or aspects of shifting, inhibition, or working memory. Bayes factors provided additional strong support in favor of no association between either measure of social isolation and any of the executive functioning components, or overall executive functions. In fact, the Bayes factors (BF_{01} = 8.70 to BF_{01} = 119.49) indicated that the data were anywhere from 8.70 to 119.49 times more likely to occur under the null hypothesis

than they were under the alternative hypotheses. Aligning with these outcomes, the corresponding effect sizes were rather small.

These findings suggest that among cognitively normal older adults, there is unlikely to be a compelling concurrent (cross-sectional) association between executive functions and subjectively experienced loneliness or objectively observed social isolation. These results are of particular interest as participants reported a range of loneliness scores, and a meaningful and variable amount of time spent alone (23.64% - 100%). Similarly, executive functioning composite z-scores were variable and captured a broad range of abilities. The lack of an association between executive functions and social interaction, therefore, is likely not attributable to restricted scores.

Although the impact of loneliness and time spent alone on executive functioning may be minimal, another possibility is that the effects in this area are transient. Zhaoyang and colleagues used mobile daily cognitive testing and found that older adults who have greater frequency of pleasant, daily social interactions with close partners, particularly friends, exhibit better cognitive performance that day and the following two days (Zhaoyang et al., 2021). In contrast, our study paired multi-day social interaction observations with static cognitive testing, which prevented assessment of potential day-to-day fluctuations in executive functions. Another recent study by Ng and colleagues using the EAR shows that the psychosocial benefit of social interaction varies by relationship status and conversation partners (Ng et al., 2022), which are factors not addressed in our study. This highlights that an important direction for future research is to pair ongoing naturalistic observation with more frequent cognitive testing, and more detailed examination of social interaction components. Importantly, naturalistic observation technology, including the EAR and similar devices, cannot capture an individual's internal mental state,

including how often they think about or desire social interaction, and whether these needs are met in their daily life.

Although our study has a number of strengths, including the objective assessment of time spent alone, the findings must be viewed in light of a few limitations, in particular related to sample characteristics and the scope of measurement. For instance, participants in this study exhibited a range of loneliness scores and a variable amount of time spent socially isolated, but they were only moderately lonely as a group. It is unclear if these results would replicate in samples of older adults who report greater feelings of loneliness, or are more isolated. A related unknown is whether our results would replicate in older adults who are not cognitively healthy, such as those with mild cognitive impairment or dementia. Future research also could include a more demographically representative sample of older adults. The current sample was composed of nearly all White, non-Latino/a, English-speaking individuals, was highly educated overall, and was screened for health factors that could moderate a relationship between executive functioning and social interaction. There are also alternative forms of social interaction that cannot be easily captured with purely ambient observation, such as text and other forms of messaging that may promote social connection (e.g., email). Future research should evaluate how these various other forms of less overtly observable social interaction behaviors, such as texting, and might relate to executive functioning.

Another future direction related to measurement is that the studies reported here did not include a self-reported measure of objective isolation (e.g., based on the Social Network Index). Prior work that has used the EAR in comparison to self-reported isolation finds that overall, participants' self-reported social isolation differs from their objectively captured social isolation. In a study comparing the EAR to time-use surveys, participants self-reported spending 25%-38%

of their waking time alone, while the EAR estimated 68.6% (Mehl & Pennebaker, 2003), with interpretational differences as to what "alone" means likely explaining the discrepancy. For instance, a person studying by themselves in a coffee shop would be considered "alone" by the EAR (adopting Goffman's (1981) definition of alone as being "a 'single,' a party of one, a person not in a *with*, a person unaccompanied 'socially' by others"; p. 79), but the person might not consider themselves to be alone because they are still surrounded by people, even if they are not engaging or interacting with them (Larson, 1990; Mehl & Pennebaker, 2003). Another EAR study comparing the sociability between Mexicans and Americans found that although Mexicans reported being less sociable than Americans, they behaved significantly more socially than Americans, as captured by the EAR (Ramírez-Esparza et al., 2009). Therefore, it would be informative for future research to understand the extent to which EAR-assessed time spent alone is correlated with other measures of putatively objective isolation. This may be important when including samples of participants in which alternative or covert forms of social interaction are likely, such as older adults with hearing loss.

As a future direction, longitudinal assessment is needed to determine whether the association between executive functioning and social interaction changes over time. Although cross-sectional studies are useful in providing information on interindividual differences, longitudinal studies present the opportunity to examine intraindividual change (Schaie, 2005). Longitudinal assessment also can provide additional confidence that a sample is aging normally, which is important for separating effects of early, "silent" pathology from healthy aging. Longitudinal assessment could additionally help elucidate whether changes in executive functioning precede changes to social function, or vice versa, and whether the nature of the relationship is linear, or non-linear. For instance, perhaps social interaction is initially associated

with better cognition but then worse cognition later on. A final consideration for future research is the complex relationship between social support and activities of daily life (Silverstein et al., 1996), and how this relationship may subsequently impact cognition over time. For example, instrumental activities of daily living like managing appointments/finances/medications, cooking, and shopping, are theorized to rely on executive functions such as inhibition, organization, planning, and decision-making (Bell-McGinty et al., 2002; Jefferson et al., 2006). Individuals who receive excess social support, and stop performing tasks they are capable of, might be at a cognitive disadvantage as they prematurely lose opportunities to use their executive functioning abilities.

Although our sample is large for the novel and time intensive use of the EAR, it is modest relative to some other studies of cognition and social connectedness in older adults. However, as noted, a recent meta-analysis of longitudinal studies in older adults, which included 102,035 unique participants, reached a similar conclusion to ours when it comes to estimates of effect sizes between cognition and social interaction (Evans et al., 2019). We suggest, therefore, that the equivocal findings in this literature may reflect that researchers are attempting to detect subtle relationships with unclear causality, at least among cognitively healthy older adults.

Conclusions

To build a complete understanding of how loneliness and social isolation relate to cognitive functioning, we not only need to know in what contexts and with what methods and measurements cognitive functioning relates to loneliness and social isolation, but also in what contexts and with what methods and measures it does not. This present study builds off the existing body of literature to offer more comprehensive insight into the social interactions of cognitive healthy older adults and establishes a foundation for future work. In sum, we did not

find compelling evidence that, among cognitively healthy older adults, there is a cross-sectional relationship between multiple aspects of executive functioning and either subjectively experienced loneliness or objectively observed time spent alone. These findings suggest that there might be a more nuanced, and/or smaller, association between the theoretical notion and empirical reality that social engagement is related to executive functioning. Registered Reports are one option for further clarifying the potential relation between social interaction and executive functioning in a transparent and reproducible manner (Chambers & Tzavella, 2021). Finally, this work also highlights the utility and potential of the EAR to naturalistically assess and capture objective concepts like isolation in an aging population.

Funding Details

This work was supported by Mind and Life Institute (2015-1440-Polsinelli).

Disclosure of Interest

The authors report no conflict of interest.

Data Availability Statement

The Electronically Activated Recorder (EAR) data are available upon request. The executive functioning data can be found here:

https://osf.io/8hkjz/?view_only=55271cf10928417f8f03564e7f353bcb

References

Aboulafia-Brakha, T., Christe, B., Martory, M.D., & Annoni, J.M. (2010). Theory of mind tasks and executive functions: A systematic review of group studies in neurology. *Journal of Neuropsychology, 5*, 39-55. https://doi.org/10.1348/174866410X533660

Althubaiti, A. (2016). Information bias in health research: definition, pitfalls, and adjustment methods. *Journal of Multidisciplinary Healthcare, 9*, 211-217.

Anderson, V. A., Anderson, P., Northam, E., Jacobs, R., & Catroppa, C. (2001). Development of executive functions though late childhood and adolescence in an Australian sample. *Developmental neuropsychology, 20*(1), 385-406.

Barnes, T. L., MacLeod, S., Tkatch, R., Ahuja, M., Albright, L., Schaeffer, J. A., & Yeh, Charlotte, S. (2021). Cumulative effect of loneliness and social isolation on health outcomes among older adults. *Aging and Mental Health, 26*(18), 1-8.

Bell-McGinty, S., Podell, K., Franzen, M., Baird, A.D., & Williams, M.J. (2002). Standard measres of executive function in predicting instrumental activities of daily living in older adults. *International Journal of Geriatric Psychiatry, 17*(9), 828-834.

Cacioppo, J. T. & Cacioppo, S. (2018). Loneliness in the modern age: An evolutionary theory of loneliness (ETL). *Advances in Experimental Social Psychology*, *58*, 127–197.

Chambers, C. D. & Tzavella, L. (2022). The past, present and future of Registered Reports. *Nature Human Behavior, 6*, 29-42.

Chen, H. L. (1994). Hearing in the elderly: Relation of hearing loss, loneliness, and self-esteem. *Journal of Gerontological Nursing, 20*(6), 22-28.

Cohen, S., Doyle, W. J., Skoner, D. P., Rabin, B. S., & Gwaltney, J. M. (1997). Social ties and susceptibility to the common cold. *JAMA, 277*(24), 1940-1944.

Cryanowski, J. M., Zill, N., Bode, R., Butt, Z., Kelly, M. A. R., Pilkonis, P. A., Salsman, J. M., & Cella. (2013). Assessing social support, companionship, and distress: National Institute of Health (NIH) Toolbox Adult Social Relationship Scales. *Health Psychology, 32*(3), 293-301. https://doi.org/10.1037/a0028586

De Winter, J. C., Gosling, S. D., & Potter, J. (2016). Comparing the Pearson and Spearman correlation coefficients across distributions and sample sizes: A tutorial using simulations and empirical data. *Psychological Methods*, *21*(3), 273-290.

DiNapoli, E.A., Wu, B., & Scogin, F. (2014). Social isolation and cognitive function in Appalachian older adults. *Research on Aging, 36*(2), 161-179.

de Jong Gierveld, J. & Havens, B. (2004). Cross-national comparisons of social isolation and loneliness: introduction and overview. *Canadian Journal on Aging, 23*(2), 109-113.

Delis, D., Kaplan, E., & Kramer, N. (2001). *Delis–Kaplan executive function system*. Odessa, FL: Psychological Assessment

Diamond, A. (2013). Executive Functions. *Annual Review of Psychology, 64*, 135-168. 10.1146/annurev-psych-113011-143750

Dykstra, P. A., van Tilburg, T. G., & de Jong Gierveld, J. (2005). Changes in older adult loneliness. *Research on Aging, 27*(6), 725-747.

Evans, I. E. M., Martyr, A., Collins, R., Brayne, C., & Clare, L. (2019). Social isolation and cognitive function in later life: A systematic review and meta-analysis. *Journal of Alzheimer's Disease, 70*, 119-144.

Fratiglioni, L., Paillard-Borg, S., & Winblad, B. (2004). An active and socially integrated lifestyle in late life might protect against dementia. *The Lancet. Neurology*, *3*(6), 343–353. https://doi.org/10.1016/S1474-4422(04)00767-7

Gilmour, H. (2011). Cognitive performance of Canadian seniors. *Health Reports, 22*, 27-31.

Goffman, E. (1981). *Forms of talk*. University of Pennsylvania Press.

Golden, C. J. (1978). *Stroop Color and Word Test: A Manual for Clinical and Experimental Uses* (pp. 1-32). Chicago, IL: Skoelting.

Griffin, S. C., Mezuk, B., Williams, A.B., Perrin, P. B., & Rybarczyk, B. R. (2020). Isolation, not loneliness or cynical hostility, predicts cognitive decline in older Americans. *Journal of Aging and Health, 32*(1-2), 52-60.

Jefferson, A.L., Paul, R.H., Ozonoff, A., & Cohen, R.A. (2006). Evaluating elements of executive functioning as predictors of instrumental activities of daily living (IADLs). *Arch Clin Neuropsychol, 21*(4), 311-320. doi:10.1016/j.acn.2006.03.007.

Kaplan, D. M., Rentscher, K. E., Lim, M., Reyes, R., Keating, D., Romero, J., ... & Mehl, M. R. (2020). Best practices for Electronically Activated Recorder (EAR) research: A practical guide to coding and processing EAR data. *Behavior Research Methods, 52*(4), 1538-1551.

Keysers, C., Gazzola, V., & Wagenmakers, E. (2021). Using Bayes factor hypothesis testing in neuroscience to establish evidence of absence. *Nature neuroscience, 23*(7), 788-799. https://doi.org/10.1038/s41593-020-0660-4

Krendl, A. C., Kennedy, D. P., Hugenberg, K., & Perry, B. L. (2022). Social cognitive abilities predict unique aspects of older adults' personal social networks. *The Journals of Gerontology, Series B: Psychological Sciences and Social Sciences, 77*(1), 18–28. doi:10.1093/geronb/gbab048

Krueger, K. R., Wilson, R. S., Kamenetsky, J. M., Barnes, L. L., Bienias, J. L., & Bennet, D. A. (2009). Social engagement and cognitive function in old age. *Experimental Aging Research, 35,* 45-60.

Lakens, D. (2022). Sample size justification. *Collabra: Psychology, 8*(1). https://doi.org/10.1525/collabra.33267

Larson, R. (1990). The solitary side of life: An explanation of the time people spend alone from childhood to old age. *Developmental Review, 10*, 155–183.

Latkin, C. A., Edwards, C., Davey-Rothwell, M. A., & Tobin, K. E. (2017). The relationship between social desirability bias and self-reports of health, substance use, and social network factors among urban substance users in Baltimore, Maryland. *Addictive behaviors*, *73*, 133–136. https://doi.org/10.1016/j.addbeh.2017.05.005

Madjar, N., North, E.A., & Karakus, M. (2019). The mediating role of perceived peer motivational climate between classroom mastery goal structure and social goal orientations. *Learning and Individual Differences, 73*, 112-123.

Mehl, M. R., Pennebaker, J. W., Crow, D. M., Dabbs, J., & Price, J. H. (2001). The electronically activated recorder (EAR): A device for sampling naturalistic daily activities and conversations. *Behavioral Research Methods, Instruments, & Computers, 33*(4), 517- 523.

Mehl, M. R., & Pennebaker, J. W. (2003). The sounds of social life: a psychometric analysis of students' daily social environments and natural conversations. *Journal of personality and social psychology*, *84*(4), 857-870.

Mehl, M. R. (2017). The electronically activated recorder (EAR): A method for the naturalistic observation of daily social behavior. *Current Directions in Psychological Science, 26*(2), 184-190. https://dx.doi.org/10.1177%2F0963721416680611

Milek, A., Butler, E. A., Tackman, A. M., Kaplan, D. M., Raison, C. L., Sbarra, D. A., ... & Mehl, M. R. (2018). "Eavesdropping on happiness" revisited: A pooled, multisample replication of the association between life satisfaction and observed daily conversation quantity and quality. *Psychological science*, *29*(9), 1451-1462.

Miyake, A., Friedman, N. P., Emerson, M. J., Witzki, A. H., Howerter, A., & Wager, T. D. (2000). The unity and diversity of executive functions and their contributions to complex "frontal lobe" tasks: A latent variable analysis. *Cognitive Psychology*, *41*(1), 49-100. doi:10.1006/cogp.1999.0734

Miyake, A. & Friedman, N. P. (2012). The nature and organization of individual differences in executive functions: Four general conclusions. *Current Directions in Psychological Science, 21*(1), 8-14. DOI: 10.1177/0963721411429458

Montoliu, T., Hidalgo, V., & Salvador, A. (2019). The relationship between loneliness and cognition in health older men and women: The role of cortisol. *Psychoneuroendocrinology, 107*, 270-279.

Morris, N., & Jones, D. M. (1990). Memory updating in working memory: The role of the central executive. *British Journal of Psychology, 81*, 111–121.

Moseley, S. (2018). Cognitive and psychosocial associations of hearing loss in older adults. Retrieved from https://repository.arizona.edu/handle/10150/628455

Mejia, S., Pineda, D., Alvarez, L., & Ardila, A. (1998). Individual differences in memory and executive function abilities during normal aging. *International Journal of Neuroscience, 95*, 271–284.

National Academies of Sciences, Engineering, and Medicine (NASEM) (2020). *Social Isolation and Loneliness in Older Adults: Opportunities for the Health Care System.* The National Academies Press. https://doi.org/10.17226/25663

Newall, N. E. G., & Menec, V. H. (2019). Loneliness and social isolation of older adults: Why it is important to examine these social aspects together. *Journal of Social and Personal Relationships, 36*(3), 925–939. https://doi.org/10.1177/0265407517749045

Ng, Y. T., Huo, M., Han, S. H., Birditt, K. S., & Fingerman, K. L. (2022). Older adult's marital status, conversation frequency, and well-being in everyday life. *The Journals of Gerontology: Series B, 77*(3), 499-512.

O'Luanaigh, C., O'Connell, H., Chin, A-V., Hamilton, F., Coen, R., Walsh, C., Walsh, J. B., … & Lawlor, B.A. (2012). Loneliness and cognition in older people: the Dublin Healthy Ageing study. *Aging & Mental Health, 16*(3), 347-352.

Otsuka, Y., Shizawa, M., Sato, A., & Itakura, S. (2021). The role of executive functions in older adults' affect theory of mind. *Archives of Gerontology and Geriatrics, 97*. https://doi.org/10.1016/j.archger.2021.104513.

Perissinotto, C. M., Cenzer, I. S., & Covinsky, K. E. (2012). Loneliness in older persons: A predictor of functional death and decline. *Archives of Internal Medicine, 172*(14), 1078-1083.

Perlman, D. & Peplau, L. A. (1998). Loneliness. In H. Friedman (Ed.), *Encyclopedia of mental health* (Vol. 2, pp. 571–581). San Diego, CA: Academic Press.

Perry, R. E., Braren, S. H., Rincón-Cortés, M., Brandes-Aitken, A. N., Chopra, D., Opendak, M., Alberini, C. M., … & Blair, C. (2019). Enhancing executive functions through social interactions: Causal evidence using a cross-species model. *Frontiers in Psychology, 10*, 2472. https://doi.org/10.3389/fpsyg.2019.02472

Polsinelli, A. (2017). Cognitive and emotional associations of mindfulness in older adults. Retrieved from https://repository.arizona.edu/handle/10150/625561

Polsinelli, A. J., Moseley, S. A., Grilli, M. D., Glisky, E. L., & Mehl, M. R. (2020). Natural, everyday language use provides a window into the integrity of older adults' executive functioning. *The Journals of Gerontology: Series B, 75*(9), e215-e220.

Premack, D. G., & Woodruff, G. (1978). Does the chimpanzee have a theory of mind? *Behavioral and Brain Sciences, 1*, 515–526. doi: 10.1017/S0140525X00076512

Raina, P., Torrance-Rynard., V., Wong, M., & Woodward, C. (2002). Agreement between self-reported and routinely collected health-care utilization data among seniors. *Health Services Research 37*(3), 751-774.

Ramírez-Esparza, N., Mehl, M. R., Álvarez-Bermúdez, J., & Pennebaker, J. W. (2009). Are Mexicans more or less sociable than Americans? Insights from a naturalistic observation study. *Journal of Research in Personality, 43*, 1-7. https://doi.org/10.1016/j.jrp.2008.09.002

Roca, M. (2016).The relationship between executive functions and theory of mind: A long and winding road. *Journal of Neurology, Neurosurgery & Psychiatry*, 87(3), 229.

Rogers, R. D., & Monsell, S. (1995). Costs of a predictable switch between simple cognitive tasks. *Journal of Experimental Psychology: General, 124*, 207–231.

RStudio Team (2021). RStudio: Integrated Development for R. RStudio, PBC, Boston, MA, URL http://www.rstudio.com/

Schaie, K. W. (2005). What can we learn from longitudinal studies of adult development? *Research in Human Development*, *2*(3), 133-158.

Schneider, W., Eschman, A., & Zuccolotto, A. (2002). E-Prime computer software and manual. Pittsburgh: Psychology Software Tools Inc.

Schnittger, R., Wherton, J., Prendergast, D., and Lawlor, B. (2012). Risk factors and mediating pathways of loneliness and social support in community-dwelling older adults. *Aging and Mental Health*, *16*, 335–346.

Schönbrodt, F. D. and Wagenmakers, E. J. (2018) Bayes factor design analysis: Planning for compelling evidence. *Psychonomic Bulletin & Review, 25*, 128–142. https://doi.org/10.3758/s13423-017-1230-y

Shankar, A., Hamer, M., McMunn, A., & Steptoe, A. (2013). Social isolation and loneliness: Relationships with cognitive function during 4 years of follow-up in the English Longitudinal Study of Ageing. *Psychosomatic Medicine, 75*, 161-170.

Silverstein, M., Chen, X., & Heller, K. (1996). Too much of a good thing? Intergenerational social support and the psychological well-being of older parents. *Journal of Marriage and Family, 4*, 970-982. https://www.jstor.org/stable/353984.

Sung, Y., Li, L., Blake, C., Betz, J., & Lin, F. R. (2016). Association of hearing loss and loneliness in older adults. *Journal of Aging and Health*, *28*(6), 979-994.

Tackman, A. M., Baranski, E. N., Danvers, A. F., Sbarra, D. A., Raison, C. L., Moseley, S. A., ... & Mehl, M. R. (2020). 'Personality in its Natural Habitat' Revisited: A Pooled, Multi–

sample Examination of the Relationships between the Big Five Personality Traits and Daily Behaviour and Language Use. *European Journal of Personality, 34*(5), 753-776.

Tourangeau, R. & Yan, T. (2007). Sensitive questions in surveys. *Psychological Bulletin, 133*(5), 859-883. https://doi.org/10.1037/0033-2909.133.5.859

Wagenmakers, E., Marsman, M., Jamil, T., Ly, A., Verhagen, J., Love, J., … Morey, R. D. (2018). Bayesian inference for psychology. Part 1: Theoretical advantages and practical ramifications. *Psychonomic bulletin & review, 25*(1), 35–57. https://doi.org/10.3758/s13423-017-1343-3

Wascher, C.A.F., Kulachi, I. G., Langley, E, & Shaw, R. C. (2018). How does cognition shape social relationships? *Phil. Trans. R. Soc. B., 373*, 1-9. http://dx.doi.org/10.1098/rstb.2017.0293

Wilson, R. S., Krueger, K. R., Arnold, S. E., Schneider, J. A., Kelly, J. F., Barnes, L. L., ... Bennett, D.A. (2007). Loneliness and risk of Alzheimer Disease. *Archives of General Psychiatry, 64*, 234-240.

von Hippel, W. (2007). Aging, executive function, and social control. *Current Directions in Psychological Science, 16*(5), 240-244. https://doi.org/10.1111/j.1467-8721.2007.00512.x

Yntema, D. B. (1963). Keeping track of several things at once. *Human Factors, 5,* 7–17.

Zhaoyang, R., Scott, S. B., Martire, L. M., & Sliwinski, M. J. (2021). Daily social interactions related to daily performance on mobile cognitive tests among older adults. *PLoS ONE, 16*(8):e0256583. https://doi.org/10.1371/journal.pone.0256583

Zhong, B., Chen, S., Tu, X., & Conwell, Y. (2017). Loneliness and cognitive function in older adults: Findings from the Chinese Longitudinal Health Longevity Survey. *J Gerontol B Psychol Sci Soc Sci, 72*(1), 120-128. doi:10.1093/geronb/gbw037